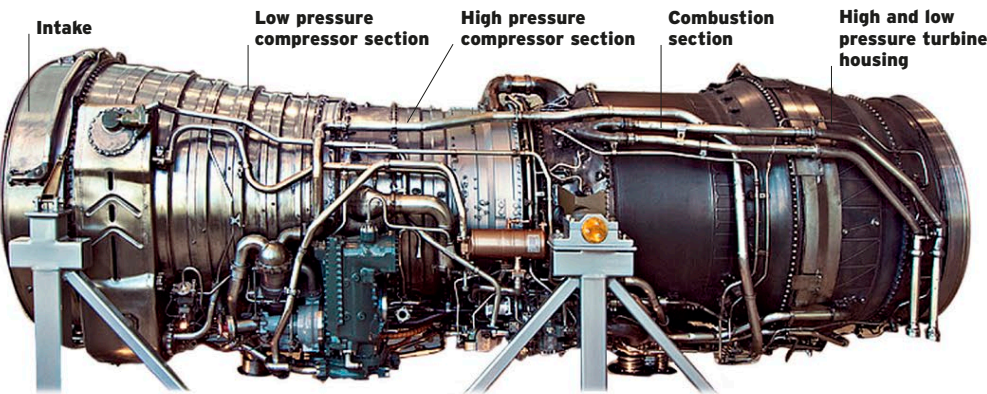
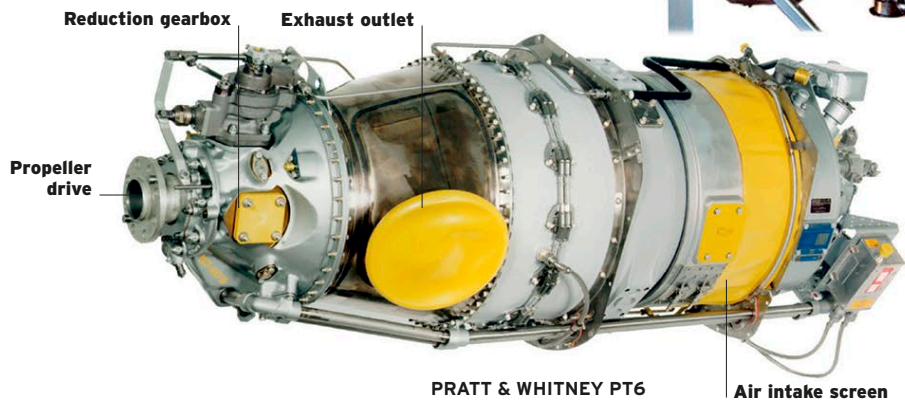


JET PROPULSION

The Whittle engine used a deceptively simple-looking centrifugal flow compressor of the type found in contemporary piston engine superchargers. It took decades to develop axial-flow engines, where air is compressed in small increments through a series of bladed compressor wheels separated by flow-aligning “stators,” to the same level of efficiency and reliability. Today, large jet engines are all of the axial-flow turbofan type, civil passenger jets using high bypass for efficiency and low noise, military types using afterburners for maximum thrust.



ROLLS ROYCE/SNECMA OLYMPUS S93



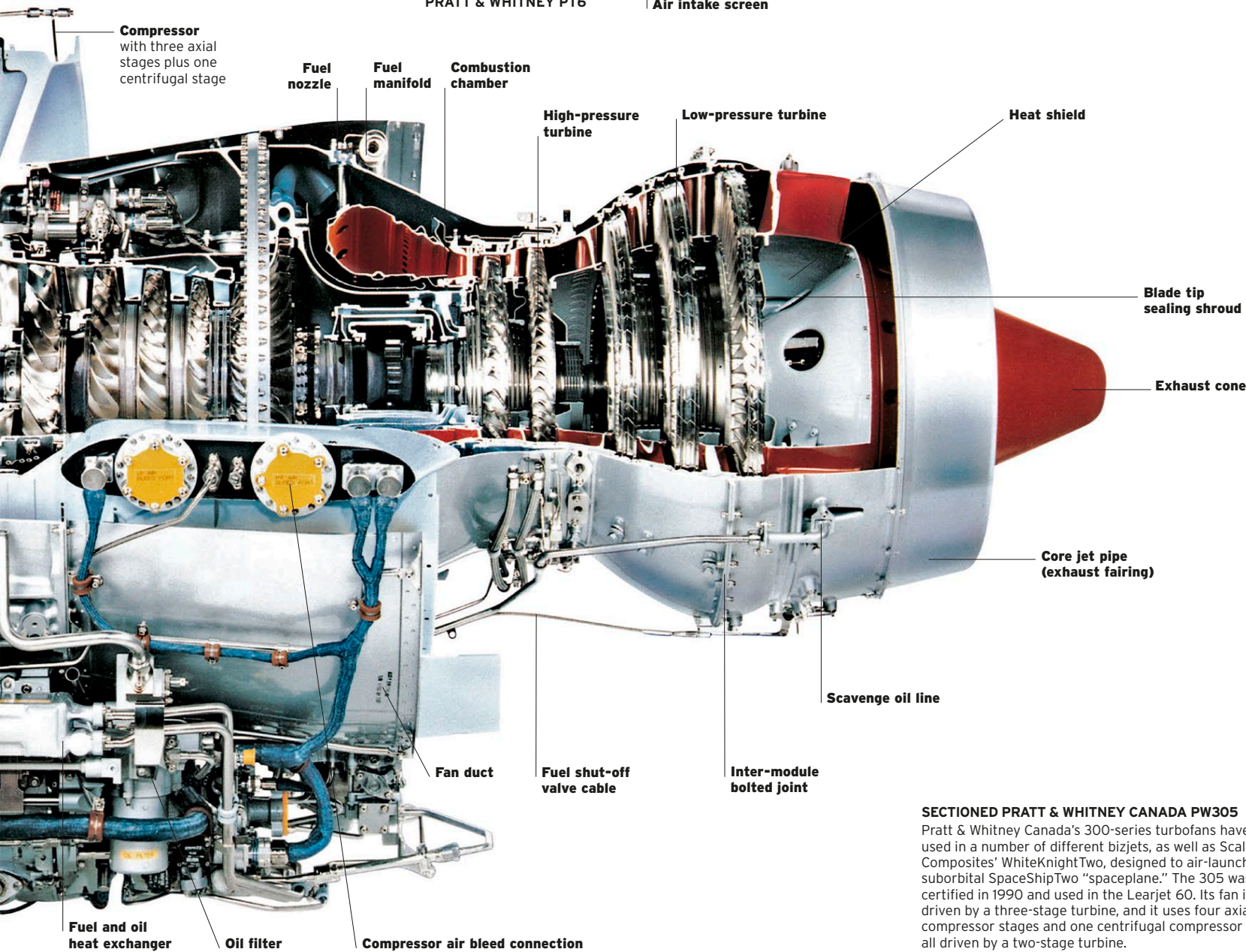
PRATT & WHITNEY PT6

◁ RELIABLE TURBOPROP

Turboprops have displaced piston engines for most propeller aircraft power units of more than 300 hp, not least for their reliability. The Pratt & Whitney Canada PT6 is an outstanding example. Available in a number of guises, it produces 600 to 2,000 hp.

△ POWER FOR CONCORDE

The Rolls-Royce/Snecma Olympus 593 was the only civil jet engine with an afterburner. Developed from the twin-spool (two compressor wheel/turbine sets) Bristol turbojet of the same name, it was only truly efficient at supersonic speed.



SECTIONED PRATT & WHITNEY CANADA PW305

Pratt & Whitney Canada's 300-series turboprops have been used in a number of different bizjets, as well as Scaled Composites' WhiteKnightTwo, designed to air-launch the suborbital SpaceShipTwo "spaceplane." The 305 was certified in 1990 and used in the Learjet 60. Its fan is driven by a three-stage turbine, and it uses four axial compressor stages and one centrifugal compressor stage, all driven by a two-stage turbine.